SIMATS ENGINEERING
CO1 - ASSESSMENT TOOL 1

COURSE NAME:Deep Learning COURSE CODE: MLA0403

Name of the Student	**U. Somesh Kumar**
Reg. No	**192425019**

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%
Assessment Tool Weightage: 35%
Duration: 1 Hour
Marks: 50

Duration: 1 Hour
QUESTIONS: 5 Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Ans: Given outcomes:
H H T H H H T H H T

- Total tosses, n = 10
- Number of Heads, H = 7

MLE Formula:

$$p\hat{}= \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$p\hat{}= \frac{7}{10} = 0.7$$

Answer: p = 0.7 (70%)

2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Ans: Students Passing an Exam

- Total students = 50
- Passed = 42

MLE Formula:

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer: Probability of passing = 0.84 (84%)

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.

Ans: Defective Light Bulbs

- Total bulbs = 100
- Defective bulbs = 8

MLE Formula:

$$\hat{p} = \frac{8}{100} = 0.08$$

Answer: Probability of a bulb being defective = 0.08 (8%)

4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.

Ans: Biased Die

- Total rolls = 60
- Number of times 6 appears = 18

MLE Formula:

$$\hat{p} = \frac{18}{60} = 0.3$$

Answer: Probability of rolling a 6 = 0.3 (30%)

5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   y = w × x
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Ans: Given:

- Input, $x = 4$

- Weight, $w = 2$
- Actual Output = 10
- Prediction:

$$y = w \times x$$

- Gradient = 4
- Learning Rate = 0.01

(1) Predicted Output

$$y = 2 \times 4 = 8$$

Answer: Predicted Output = 8

(2) Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$
$$= 10 - 8 = 2$$

Answer: Error = 2

(3) Updated Weight

Weight update formula:

$$w_{\text{new}} = w - (\eta \times \text{gradient})$$

where

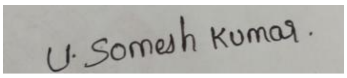
- $w = 2$
- $\eta = 0.01$
- Gradient = 4

$$w_{\text{new}} = 2 - (0.01 \times 4)$$
$$= 2 - 0.04 = 1.96$$

Answer: Updated Weight = 1.96

Code of Conduct:

*I **U. Somesh Kumar (192425019)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.



U. Somesh Kumar

Signature of the student: *Name of the student:*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation